Hudson Heuter

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Education

Boston University | Mechanical Engineering B.S. (09/2025 - Present)

- **Cumulative GPA:** 3.88 / 4.0
- **Activities:** Sophomore Chair of Engineering Student Government, Delta Tau Delta Executive Board, BU Rocket Propulsion Group, BU Instep, BU UAV

Irvington High School (09/2021 - 05/2025)

- **Activities:** Section Champion Track, All-County High Jump, Robotics Club, American Computer Science League

SAT: Composite: 1550 | Reading and Writing: 760 | Math: 790

Personal Projects

Automated Card Shuffling Mechanism **May 2026 – Aug 2026**

- Modeled mechanism using SolidWorks, refining the design through rounds of test-prints
- Integrated a motor-driven roller system to automatically and consistently riffle two halves of a deck together
- Achieved single-card dispensing across ~70% of shuffle cycles

BU Rocket Propulsion Group Intro Project **Jan 2026 – May 2026**

- Designed and analyzed a rocket engine nozzle as part of a 4-person team
- Developed detailed CAD models, part drawings, and FEA (Finite Element Analysis) plots, incorporating design reviews, design rationale, and manufacturability considerations to reach a test-ready configuration

Room Temperature Monitor **Jan 2026 – Apr 2026**

- Built real-time temperature monitoring system with circuit design, wiring, soldering, and power management

Grid-Aware AI Training Scheduler **Jan 2026 – Mar 2026**

- Built a prototyped web app forecasting optimal AI training windows using live CAISO energy grid data
- Identified optimal windows with ~60% lower electricity cost and ~85% lower grid net load compared to daily peak, using live CAISO pricing and generation data
- https://caiso-ai-train.onrender.com/

Experience

BURPG | Structures Team Member | Boston, MA **Feb 2025 - Present**

- Programmed CAM toolpaths and machined launch lugs and actuator adapter for rocket engine on a CNC mill
- Assisted in drilling mounting holes for rocket fins with a drill press

Goosefeather | Busser | Irvington, NY **Apr 2023 – Aug 2026**

- Delivered consistent, high-quality guest experiences at Michelin-guide restaurant through clear communication
- Demonstrated reliability in a fast-paced environment, balancing multiple priorities and maintaining professionalism

Tutor | Irvington, NY **Nov 2022 – Jun 2023**

- Provided one-on-one and small-group instruction across many subjects for students in multiple grade levels

FoodShareJoy | Co-Founder | Irvington, NY **Feb 2022 – May 2025**

- Student-run nonprofit organization; created 8,350 total meals

Technical Skills & Additional Information

Engineering: SolidWorks, 3D CAD modeling, machining, CAM, engineering drawings, FEA, design reviews, DFM, prototyping, 3D Printing, Python, debugging

Certifications: Google IT Automation with Python Specialization

Interests: Dungeons & Dragons, Biking, Skateboarding, Chess, Pickup Basketball, Videogame Programming